AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational Databases

Final Examination
Spring Semester 2025/2026

Date: 2 Jun 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: 50
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I [10 Marks]

This part includes 10 MCQ; each weights 1 mark. Choose the most suitable answer for the following question

1- Which SQL command is used to remove all rows from a table but keep its structure?
   a. DELETE
   b. DROP
   c. TRUNCATE
   d. ALTER

2- Which of the following is *not* a requirement for a relation to be in First Normal Form (1NF)?
   a. All attributes must have atomic values
   b. There should be no repeating groups
   c. It must have a composite primary key
   d. Each column should contain values of the same type

3- If a relation is in 1NF and all non-key attributes are fully functionally dependent on the whole primary key, the relation is in:
   a. First Normal Form
   b. Second Normal Form
   c. Third Normal Form
   d. Boyce-Codd Normal Form

4- Which of the following functional dependencies violates 2NF in the relation patient_prescription(patient_id, date, drug_code, drug_name)?
   a. drug_code → drug_name
   b. patient_id, date → drug_code
   c. patient_id → date
   d. drug_code → patient_id

5- Which of the following statements best describes a transitive dependency?
   a. An attribute depends on part of a composite key
   b. An attribute is determined by a non-key attribute
   c. An attribute determines the primary key
   d. An attribute repeats within the table

6- Consider the relation:
   student_enrollment(StudentID, StudentName, CourseID, CourseName, InstructorName, Department)
   a. Which of the following is the correct approach to convert it to 3NF?
   b. Remove repeating groups only
   c. Remove attributes that depend on the primary key partially
   d. Remove attributes that depend on non-primary key attributes into separate relations
   e. Merge all into a single relation with a composite key

7- Which of the following anomalies is most likely to occur in a relation that is in 1NF but not 2NF?
   a. A. Insertion anomaly due to null values
   b. Deletion of a tuple causes loss of independent data
   c. Redundant data caused by partial dependencies
   d. Loss of functional dependencies

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8- A relation R(A, B, C, D) has the following dependencies:
AB → C
C → D
Which of the following is a candidate key?

a. AB
b. C
c. ABC
d. ABD

9- Which of the following best describes a lossless-join decomposition?

a. A decomposition that prevents any null values
b. A decomposition where no data is lost when relations are recombined
c. A decomposition where all transitive dependencies are removed
d. A decomposition that results in 2NF

10- Which of the following relations is in Boyce-Codd Normal Form (BCNF)?

a. A relation where every determinant is a candidate key
b. A relation with no multivalued attributes
c. A relation in 3NF where transitive dependencies exist
d. A relation that is decomposed into smaller relations with fewer attributes

Part II [40 Marks]

Question 2 [20 Marks]

Consider the following schema for an online store:

- Researcher(ResearcherID, Name, Department, Rank)
- Project(ProjectID, Title, StartDate, EndDate, FundingAmount)
- Publication(PubID, Title, Year, ResearcherID)
- Collaboration(ProjectID, ResearcherID, Role)
- FundingAgency(AgencyID, Name, Country)
- Funding(AgencyID, ProjectID)

Write SQL queries for the following:

1. List the names of researchers who have worked on all projects funded by the agency named “Qassim Science Foundation”. **[4 marks]**
2. Find the top 2 researchers (by name) who have published the highest number of publications between 2021 and 2025. **[4 marks]**
3. Retrieve the names of all departments that have at least one researcher working on a project that received more than 500,000 SAR in funding. **[4 marks]**
4. For each funding agency, display the agency name and the total number of unique researchers involved in projects they funded. **[4 marks]**
5. List the researchers who collaborated on a project with at least 2 different roles (e.g., Analyst, Lead, Reviewer). **[4 marks]**

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Question 3 [10 Marks]

A hospital stores equipment purchase information in a single table called EquipmentPurchases with the following sample data:

PurchaseID	SupplierName	SupplierPhone	EquipmentList	PurchaseDate	TotalCost
P001	MedSupply Co	555-1234	{ECG Monitor, 1200}, {Stethoscope, 100}	2024-11-01	1300
P002	HealthTech	555-5678	{Infusion Pump, 2500}, {Thermometer, 300}, {ECG Monitor, 1200}	2024-11-03	4000
P003	MedSupply Co	555-1234	{Ventilator, 5500}	2024-11-05	5500

1. Convert the table to First Normal Form (1NF) **[3 marks]**
2. Convert to Second Normal Form (2NF) **[3 marks]**
3. Convert to Third Normal Form (3NF) **[4 marks]**

Question 4 [10 Marks]

A university's database administrator is managing access to a StudentRecords database. The DBA wants to grant and restrict privileges to certain users to ensure privacy and integrity of student data.

Answer the following questions using your understanding of SQL privileges and DBA professional responsibilities.

1. Write the SQL statement to grant user "Lina" the ability to view student grades from the Grades table but prevent her from making any updates. **[2 marks]**
2. Revoke the ability to insert new records into the Enrollments table from user "Tariq". **[2 marks]**
3. Explain, in your own words, what the principle of "Integrity" means in the context of the DBA Code of Ethics. **[2 marks]**
4. Give one real-world example of how a DBA can apply the principle of "Confidentiality" while managing access to a healthcare database. **[2 marks]**
5. What is the risk of using the GRANT OPTION when allowing teaching assistants access to student data in a shared academic database? **[2 marks]**

End of questions

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